

Chapter 5

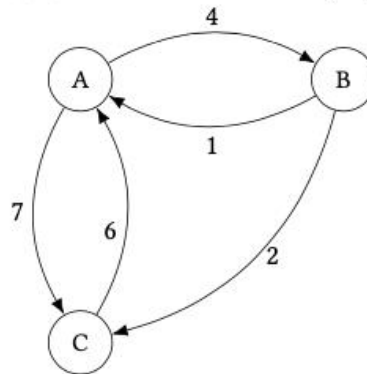
Dynamic Programming

1. Define Dynamic Programming. Explain its properties.
2. Differentiate between Dynamic Programming vs Greedy Algorithm.
- 3.

Question 5.1

Assignment: All-Pairs Shortest Paths

Consider a directed graph with vertices A, B, and C and the following edge weights:



Task: Compute the shortest path distances between *every pair* of vertices. Show each distance matrix after each iteration.

4. Explain Travelling Salesperson Problem (TSP) algorithm with its time and space complexity.
5. Compute the edit distance between the strings "artificial" and "Intelligence" using the dynamic programming approach. Show the DP table after filling it out.
- 6.

Question 5.2

Question 1: Solve the following 0/1 Knapsack problem using dynamic programming. Knapsack capacity $W = 4$.

Item	Weight (w_i)	Value (v_i)
I_1	1	2
I_2	2	3
I_3	3	4

Solution: Selected items $\{I_1, I_3\}$ with total weight $1 + 3 = 4$ and maximum value = 6.

7. Find the optimal parenthesization for the matrix chain product ABCD with size of each is given as $A_{5 \times 10}$, $B_{10 \times 15}$, $C_{15 \times 20}$, $D_{20 \times 30}$.